Vasundhara Prasad

Astrophysics PhD Student

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vasu-prasad

Research Interests

- Protoplanetary Discs
- Planet Formation
- Numerical Simulations

Education

PhD in Astronomy *Oct 2023 —*
Institute of Astronomy, University of Cambridge

- **Thesis:** Stellar Flybys in Protoplanetary Discs
- Supervisors: Professor Cathie Clarke & Dr Cristiano Longarini

MSci in Natural Sciences (Astrophysics) *Oct 2018 – June 2022*
University of Cambridge

- Graduated with First Class Honours
- **Master's thesis:** The first black holes and their gravitational wave signature

Publications

Kinematic Signatures of Stellar Flybys in Protoplanetary Discs *Nov 2026 (submitted)*
Vasundhara Prasad, Cristiano Longarini, Carolin Kimmig, Cathie Clarke
arXiv:2611.12543

Dust trapping in protoplanetary discs after stellar flybys *Sep 2025*
Vasundhara Prasad, Cristiano Longarini, Cathie Clarke
MNRAS, 543, 1798

Awards and Prizes

Examination Prize 2022 (£240) *Aug 2022*
Trinity College, Cambridge
Awarded in recognition of excellent performance in final examinations.

Talks

Invited Talks:

Planets Journal Club – Queen Mary University of London *Aug 2026*
Spirals and Warps: Kinematics and Dust Dynamics of Protoplanetary Discs after Stellar Flybys

Special Seminar – IPAG, Grenoble *June 2026*
Spirals and Warps: The Gas and Dust Response of Protoplanetary Discs to Stellar Flybys

Dynamics Lunch Seminar Series – Institute of Astronomy, Cambridge *Aug 2025*
Dust Trapping in Protoplanetary Discs after Stellar Flybys

Contributed Talks:

UK & Ireland Discs Meeting 2026 – DAMTP, Cambridge *Jul 2026*
Kinematic Signatures of Stellar Flybys in Protoplanetary Discs

James Williams Research Colloquium – Trinity College, Cambridge *June 2026*
How Stars Make Planets: The Effect of Stellar Flybys on Protoplanetary Discs

Multiplicity in Young Stars – Niels Bohr Institute, Copenhagen *Aug 2025*

Dust Dynamics in Protoplanetary Discs after Stellar Flybys

PHANTOM Users Workshop – IPAG, Grenoble *June 2025*

Dust Dynamics in Protoplanetary Discs after Stellar Flybys

DYNAMIX – Institute of Astronomy, Cambridge *June 2025*

Dust Dynamics in Protoplanetary Discs after Stellar Flybys

Wednesday Seminar Series – Institute of Astronomy, Cambridge *Mar 2025*

Stellar Flybys in Protoplanetary Discs

Trinity College Science Society Annual Symposium – Trinity College, Cambridge *Mar 2022*

Gravitational Waves from the First Black Holes

Experience

Quantum Computing Consultant *Sep 2022 — Sep 2023*

Data Reply UK, London

Using Python to program Quantum Annealing computers for financial use cases.

Technical Skills

Programming Languages: Python (experienced), C++ (familiar), MATLAB (familiar)

Tools and Environments: bash, git, LaTeX

OS: Linux, HPC supercomputing

Teaching

Supervisor, Part II Astrophysics, University of Cambridge *Jan 2024 –*

Conducted small-group teaching and marking solutions for the Astrophysical Fluid Dynamics lecture course.

Mathematics and Physics Tutor *Aug 2018 – June 2022*

Weekly one-to-one tutoring sessions with students preparing for GCSE Mathematics and for A Level Mathematics and Physics.

Professional Service

Organiser, Exoplanet Seminar Series *Aug 2026 –*

University of Cambridge

A long-running cross-departmental seminar series in the field of exoplanets.

Admissions and Recruitment Focus Group, Equality and Diversity Committee *Dec 2024 –*

Institute of Astronomy, University of Cambridge

Member of Admissions & Recruitment Focus Group, analysing the Institute's admissions data in order to improve diversity of admissions.

Organiser, Wednesday Seminar Series *Jan 2025 – June 2025*

Institute of Astronomy, University of Cambridge

A long-running seminar series given by PhD students and postdocs at the Institute of Astronomy.

Organiser, Undergraduate Journal Club *Oct 2023 – June 2024*

Institute of Astronomy, University of Cambridge

A journal club for undergraduates, also offering career advice.